Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/10/2005
PI Name:	Kihm, Kenneth D Ph.D.		
Project Title:	Microscale Investigation of the Thermo-Fluid Transport in the Transition Film Region of an Evaporating Capillary Meniscus Using a Microgravity Environment		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Interfacial phenomena		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	University of Tennessee		
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City:	Knoxville	**State:**	TN
Zip Code:	37996	**Congressional District:**	2
Comments:	Moved to U Tennessee from Texas A&M in 2004-2005.		
Project Type:	Flight	**Solicitation / Funding Source:**	2001 Fluid Physics 01-OBPR-02: Research and Flight Experiment Opportunities
Start Date:	10/01/2002	**End Date:**	09/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Allen, Jeffrey S. (National Center for Microgravity Research) Hallinan, Kevin P. (University of Dayton) Pratt, David M. (Wright-Patterson Air Force Base)		
Grant/Contract No.:	NNC05GA18G ; NAG3-2712		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	In order to enhance the fundamental understanding of thin film evaporation and thereby improve the critical design concept for two-phase heat transfer devices, microscale heat and mass transport is to be investigated for the transition film region using state-of-the-art optical diagnostic techniques. By utilizing a microgravity environment, the length scales of the transition film region can be extended sufficiently, from submicron to micron, to probe and measure the microscale transport fields which are affected by intermolecular forces. Extension of the thin film dimensions under microgravity will be achieved by using a conical evaporator made of a thin silicon substrate under which concentric and individually controlled micro-heaters are vapor-deposited to maintain either a constant surface temperature or a controlled temperature variation. Local heat transfer rates, required to maintain the desired wall temperature boundary condition, will be measured and recorded by the concentric thermoresistance heaters controlled by a Wheatstone bridge circuit. The proposed experiment employs a novel technique to maintain a constant liquid volume and liquid pressure in the capillary region of the evaporating meniscus so as to maintain quasi-stationary conditions during measurements on the transition film region. Alternating use of Fizeau interferometry via white and monochromatic light sources will measure the thin film slope and thickness variation, respectively. Molecular Fluorescence Tracking Velocimetry (MFTV), utilizing caged fluorophores of approximately 10-nm in size as seeding particles, will be used to measure the velocity profiles in the thin film region. An optical sectioning technique using confocal microscopy will allow submicron depthwise resolution for the velocity measurements within the film for thicknesses on the order of a few microns. Digital analysis of the fluorescence image-displacement PDFs, as described in the main proposal, can further enhance the depthwise resolution. The primary objectives of this research are: 1. Unique design and fabrication of micro-heater array that allows heat flux measurements with 100-micron spatial resolution. 2. Thorough review and improvement of the previously published theoretical models for local evaporative flux and inner scale thermo-fluid dynamics of an evaporating contact line. 3. Develop a unique, microscale flow field measurement technique by the combined use of a high-speed and optically-sliced CLSM imaging and the innovative MFTV. Numerous technologies, both terrestrial and space-based, rely upon phase change. These technologies include thermal control systems, application of films and coatings and adhesives, chemical and bioprocessing, and pharmaceutical development. The development of these technologies has relied upon the time tested method of trial and error to develop empirical correlations and phenomenological models of bulk behavior. As systems are reduced toward the micro and nano scales, the role of surface forces on system behavior will continue to become more important and there is little data upon which to build new correlations or models. In order to understand the behavior of systems dominated by surface interactions, we need quantitative data of the transport behavior in the transition film region. Understanding the transition region is critical to predicting and controlling the behavior of evaporating liquid films and menisci in microsystems. This improved fundamental understanding will aid in the refinement of design methodologies of capillary-pumped heat transport devices, thereby benefitting not only NASA missions relying upon these in thermal management and power systems, but the rapidly evolving fields of MEMS and nanotechnology.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Advances in medicine and biology hold the promise for improvements in quality of life from a health perspective. Pursuant to this promise is the rapid testing and development of pharmaceuticals. Development of new medicines is increasingly relying upon automated microfluidic technologies such as the lab-on-a-chip. As with microscale heat transfer systems, the role of surface interactions in medical applications is increasing as the length scales decrease. Though the impact of the proposed research on the medical community is far in the future, a more thorough understanding of the transport phenomena in the transition film region will improve the ability to develop new medicines.
Task Progress:	In order to measure the micro-scale flow field inside the thin evaporating film, a new particle tracking velocimetry (PTV) technique has been developed by using fluorescent nanoparticles. Quantitative tracking of nanoparticles has been achieved by using Total Internal Reflection Fluorescence Microscopy (TIRFM). The implemented system shows that 200-nm fluorescence particles can be three-dimensionally tracked: an elaborate analysis of their image intensity detemines their line-of-sight locations and velocities, and a robust algorithm for planar particle tracking specify their lateral locations and velocity components. The technic proves that nanoparticles can be quantitatively tracked within the range of a few hundred nanometers from the solid interface.
Bibliography Type:	Description: (Last Updated: 02/26/2014)
Articles in Peer-reviewed Journals	Park JS, Choi CK, Kihm KD. "Optically sliced micro-PIV using confocal laser scanning microscopy (CLSM)." Experiments in Fluids 2004 Jul;37(1):105-19. http://dx.doi.org/10.1007/s00348-004-0790-6 , Jul-2004
Dissertations and Theses	Wee SK. "Microscale observables for heat and mass transport in submicron-scale evaporating thin film." Ph.D. Dissertation, Texas A&M University, May 2004. , May-2004
Papers from Meeting Proceedings	Kihm KD, Banerjee A. "Tracking of nanoparticles using evanescent wave microscopy." ASME-Heat Transfer/Fluids Engineering Summer Conference, Charlotte, NC, July 2004. HT-FED 2004-56652 ASME-Heat Transfer/Fluids Engineering Summer Conference, Paper HT-FED 2004-56652, 2004. , Jul-2004
Presentation	Kihm, K. D. "Why and how do we track thermal behaviors of nanoparticles in micro/nano-scale resolution?" National Science Foundation Invited Presentation May-2004